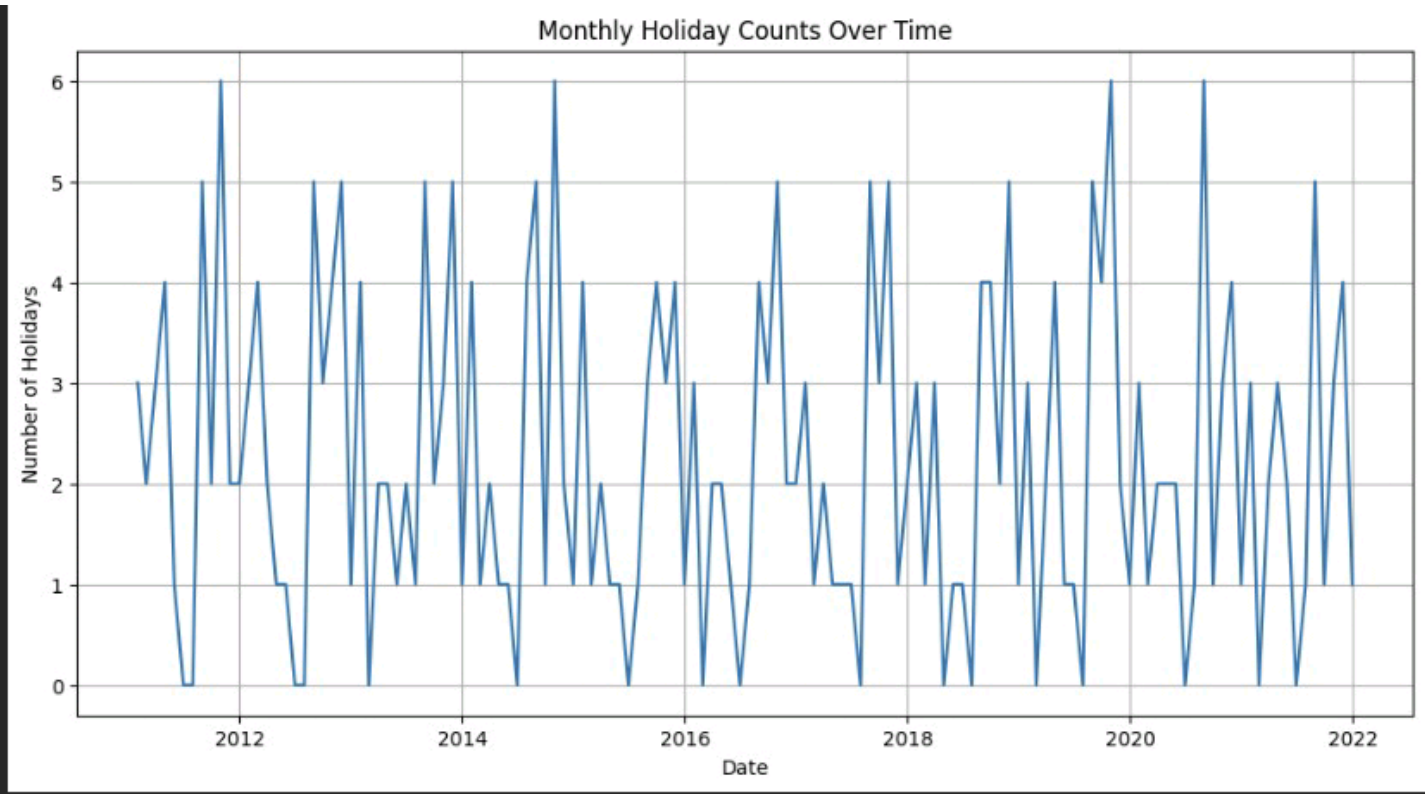
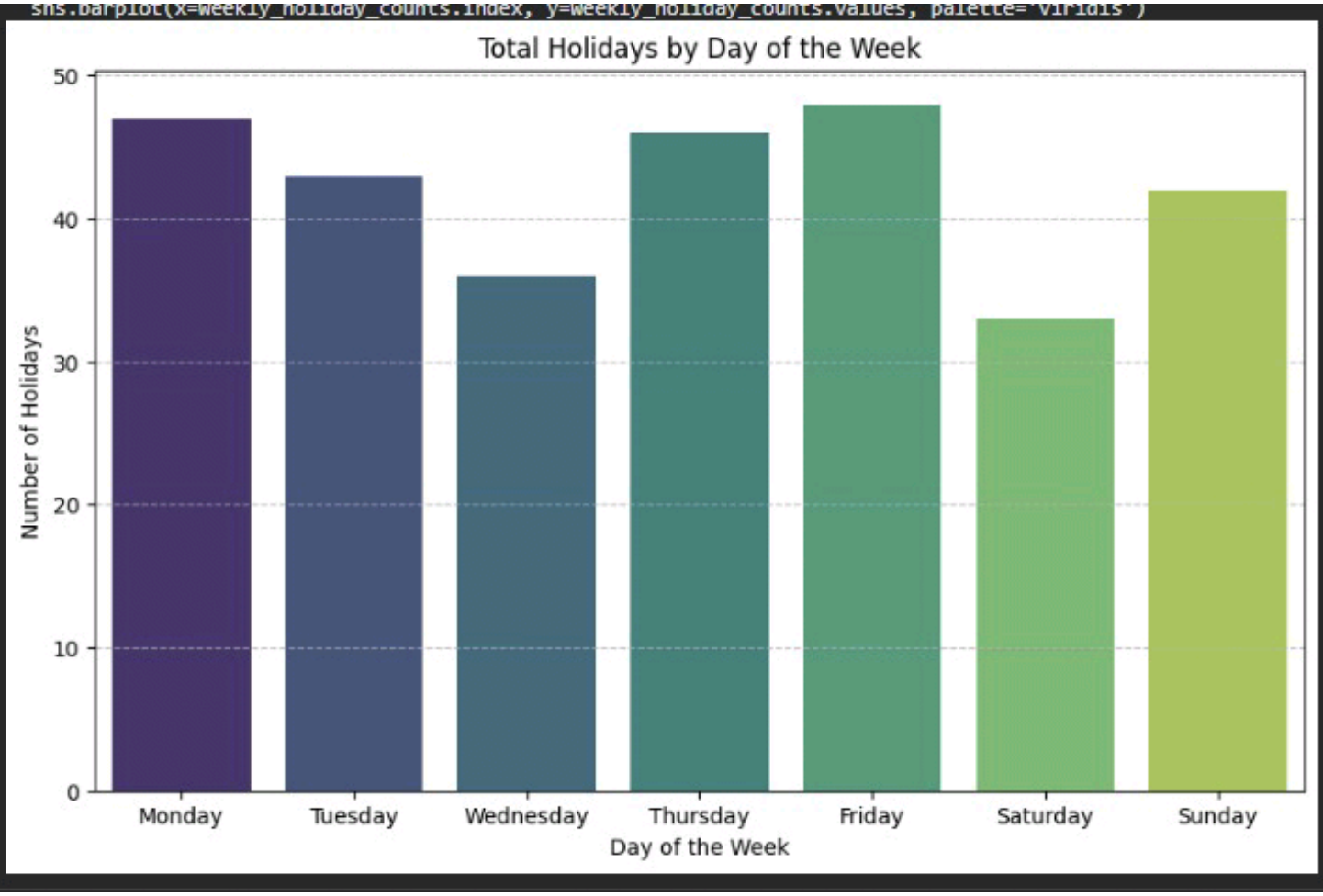
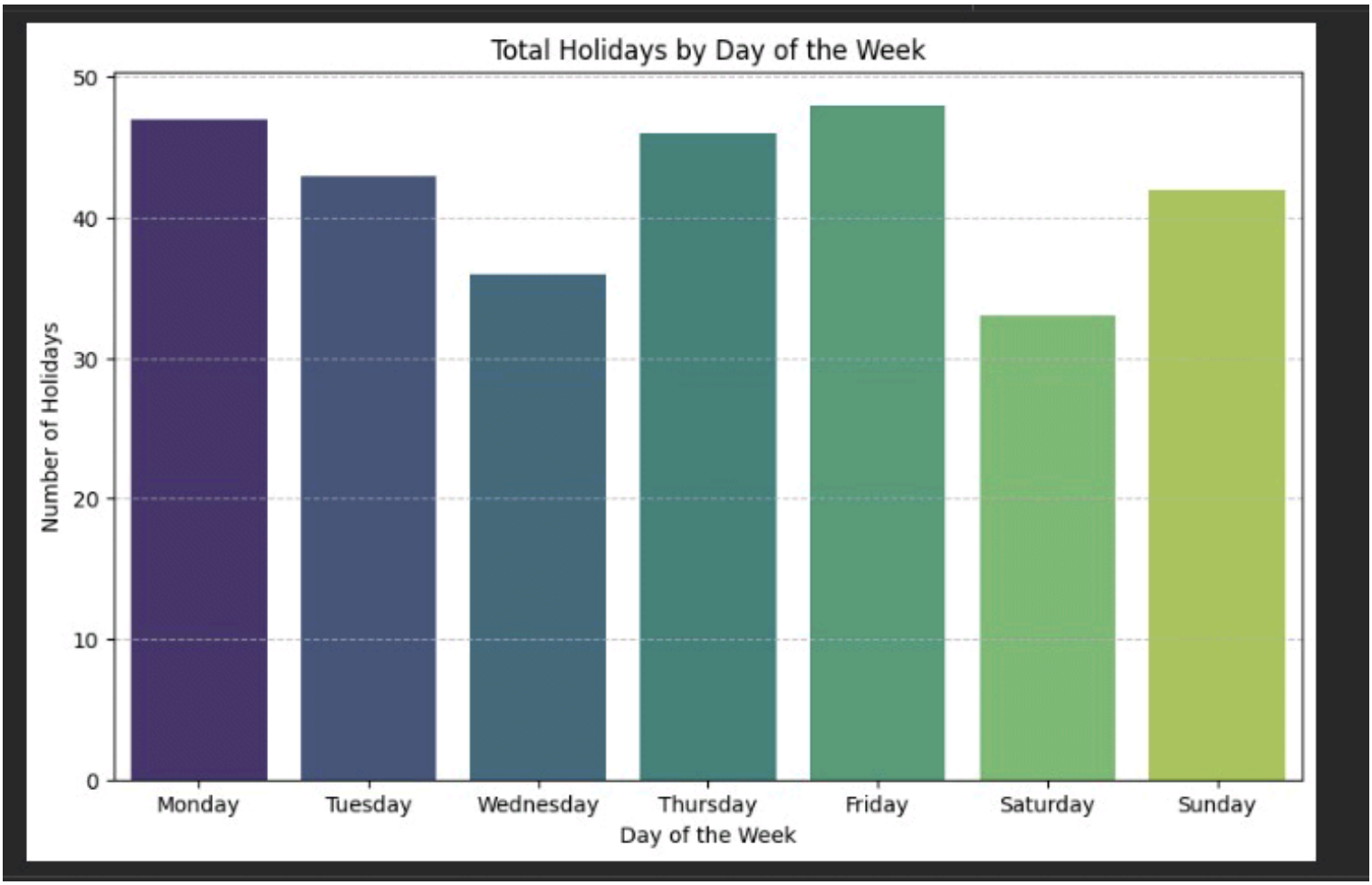
```python
test_df['penalty'] = np.where(
    test_df['actual'] > test_df['forecast'],
    4 * (test_df['actual'] - test_df['forecast']),
    2 * (test_df['forecast'] - test_df['actual'])
)

total_penalty = test_df['penalty'].sum()
print("Total Deviation Penalty ₹:", total_penalty)
```

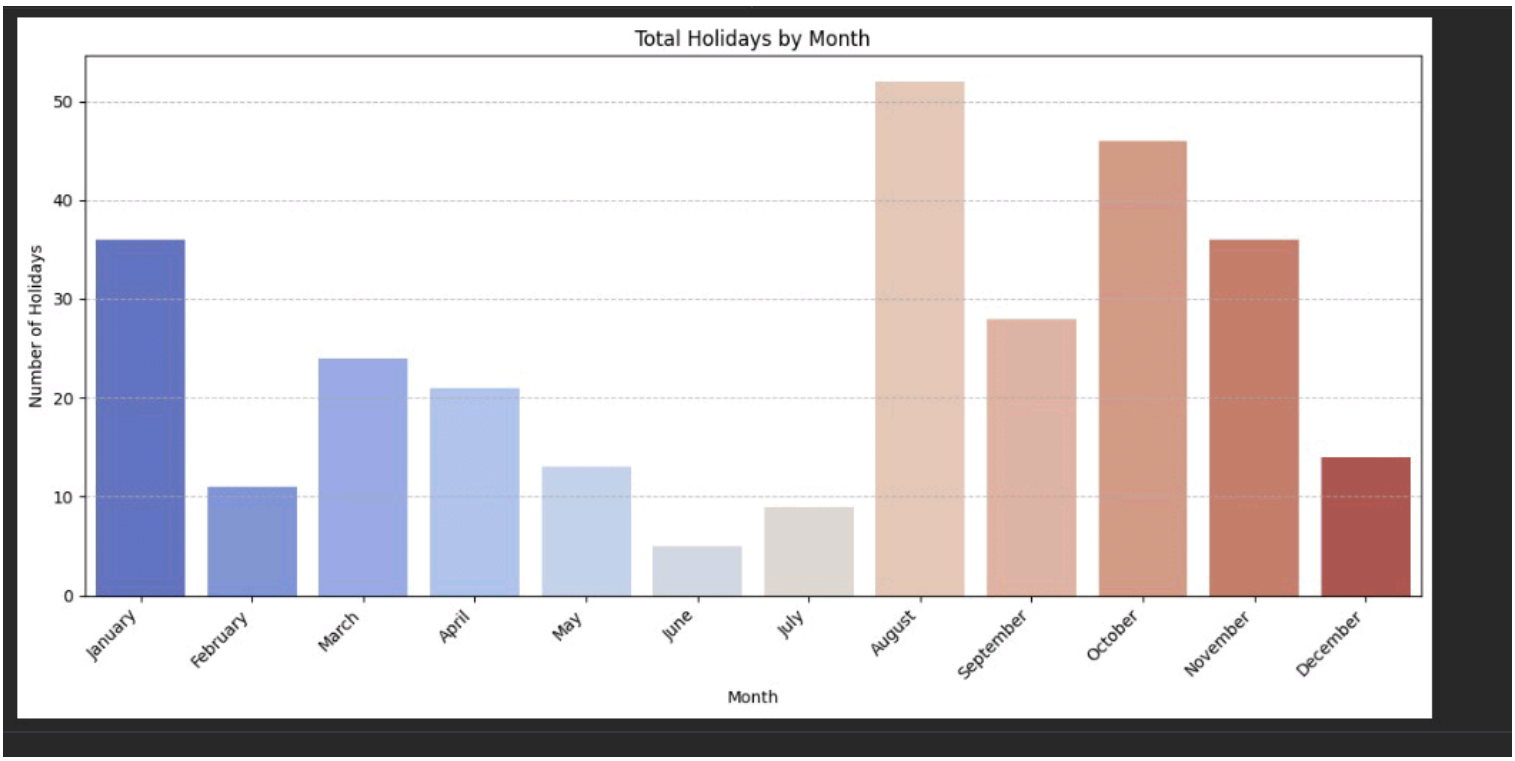
```
Total Deviation Penalty ₹: 51.58128976197779
```

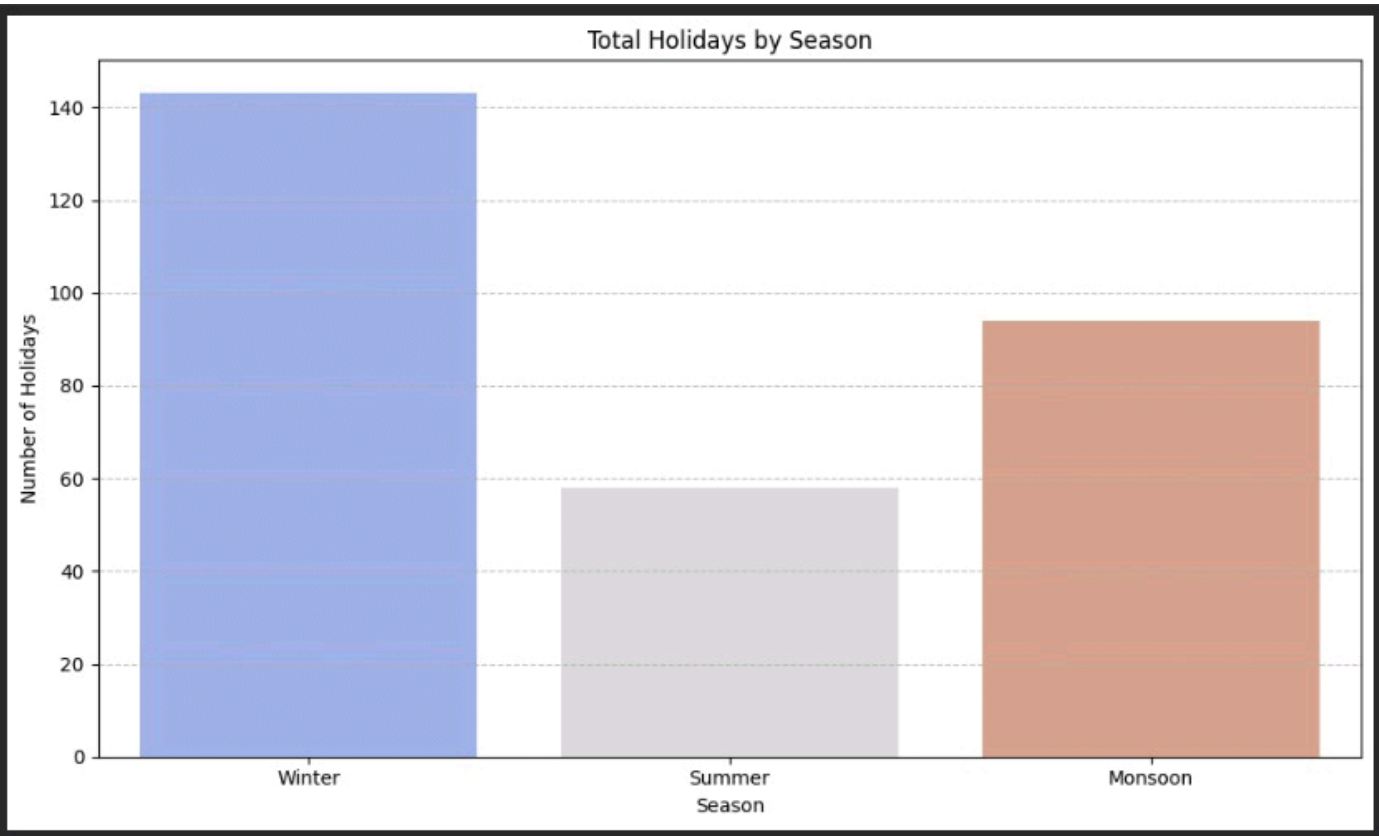

Monthly Holiday Counts Over Time
Number of Holidays
0
1
2
3
4
5
6
2012
2014
2016
2018
2020
2022
Date

```
sns.barplot(x=weekly_holiday_counts.index, y=weekly_holiday_counts.values, palette='viridis')
```




Total Holidays by Day of the Week
Number of Holidays
0
10
20
30
40
50
Monday
Tuesday
Wednesday
Thursday
Friday
Saturday
Sunday
Day of the Week


Total Holidays by Month
Number of Holidays
0
10
20
30
40
50
January
February
March
April
May
June
July
August
September
October
November
December
Month


Total Holidays by Season
Number of Holidays
0
20
40
60
80
100
120
140
Winter
Summer
Monsoon
Season

```
Feature DataFrame 'X' after one-hot encoding categorical features (head):
            Is_Event_Day  Festival_Indicator  Is_Weekend  \
Date
2011-01-01             1                   0           1
2011-01-15             1                   0           1
2011-01-26             1                   0           0
2011-02-16             1                   0           0
2011-02-19             1                   0           1

            Holiday_Ind_Lag_1_Day  Holiday_Ind_Lag_7_Day  Day_of_Week_Monday  \
Date
2011-01-01                    0.0                    0.0               False
2011-01-15                    1.0                    0.0               False
2011-01-26                    1.0                    0.0               False
2011-02-16                    1.0                    0.0               False
2011-02-19                    1.0                    0.0               False

            Day_of_Week_Saturday  Day_of_Week_Sunday  Day_of_Week_Thursday  \
Date
2011-01-01                  True               False                 False
2011-01-15                  True               False                 False
2011-01-26                 False               False                 False
2011-02-16                 False               False                 False
2011-02-19                  True               False                 False

            Day_of_Week_Tuesday  ...  Month_Name_January  Month_Name_July  \
Date                             ...
2011-01-01                False  ...               True            False
2011-01-15                False  ...               True            False
2011-01-26                False  ...               True            False
2011-02-16                False  ...              False            False
2011-02-19                False  ...              False            False

            Month_Name_June  Month_Name_March  Month_Name_May  \
Date
2011-01-01            False             False           False
2011-01-15            False             False           False
2011-01-26            False             False           False
2011-02-16            False             False           False
2011-02-19            False             False           False

            Month_Name_November  Month_Name_October  Month_Name_September  \
Date
2011-01-01                False               False                 False
2011-01-15                False               False                 False
2011-01-26                False               False                 False
2011-02-16                False               False                 False
2011-02-19                False               False                 False

            Season_Summer  Season_Winter
Date
2011-01-01          False           True
2011-01-15          False           True
2011-01-26          False           True
2011-02-16          False           True
2011-02-19          False           True

[5 rows x 24 columns]
```

	date	event_name	holiday_ind	remarks	Is_Event_Day	Festival_Indicator	Day_of_Week	Month_Name	Season	Is_Weekend	Holiday_Ind_Lag_1_Day	Holiday_Ind_Lag_7_Day	Holiday_Ind_Rolling_Mean_7_Day	date_parsed
37	2011-10-02	Navratri Day5	0	NaN	1	1	Sunday	October	Winter	1	0.0	0.0	0.142857	2011-10-02
38	2011-10-02	Gandhi Jayanti	1	NaN	1	0	Sunday	October	Winter	1	0.0	0.0	0.285714	2011-10-02
148	2013-09-15	Ganesh Festival Special Day7	0	NaN	1	0	Sunday	September	Monsoon	1	0.0	1.0	0.142857	2013-09-15
149	2013-09-15	Heavy Rain	0	NaN	1	0	Sunday	September	Monsoon	1	0.0	1.0	0.000000	2013-09-15
175	2014-01-14	Id-e-Milad	1	NaN	1	0	Tuesday	January	Winter	0	1.0	1.0	0.857143	2014-01-14
176	2014-01-14	Makar Sankranti	1	NaN	1	0	Tuesday	January	Winter	0	1.0	1.0	0.857143	2014-01-14
199	2014-07-29	Heavy Rain	0	NaN	1	0	Tuesday	July	Monsoon	0	0.0	0.0	0.285714	2014-07-29
200	2014-07-29	Ramzan ID	1	NaN	1	0	Tuesday	July	Monsoon	0	0.0	0.0	0.428571	2014-07-29
204	2014-08-18	Janmashtmi	1	NaN	1	0	Monday	August	Monsoon	0	1.0	1.0	0.571429	2014-08-18
205	2014-08-18	Parsi New Year	1	NaN	1	0	Monday	August	Monsoon	0	1.0	0.0	0.714286	2014-08-18
208	2014-08-31	Ganesh Festival Day3	0	NaN	1	0	Sunday	August	Monsoon	1	0.0	0.0	0.714286	2014-08-31
209	2014-08-31	Heavy Rain	0	NaN	1	0	Sunday	August	Monsoon	1	0.0	1.0	0.571429	2014-08-31
211	2014-09-01	Heavy Rain	0	NaN	1	0	Monday	September	Monsoon	0	0.0	1.0	0.285714	2014-09-01
210	2014-09-01	Ganesh Festival Day4	0	NaN	1	0	Monday	September	Monsoon	0	0.0	1.0	0.428571	2014-09-01
212	2014-09-02	Ganesh Festival Special Day5	0	NaN	1	0	Tuesday	September	Monsoon	0	0.0	1.0	0.142857	2014-09-02
213	2014-09-02	Grid Disturbance	0	NaN	1	0	Tuesday	September	Monsoon	0	0.0	1.0	0.000000	2014-09-02
227	2014-10-02	Gandhi Jayanti	1	NaN	1	0	Thursday	October	Winter	0	0.0	0.0	0.142857	2014-10-02
228	2014-10-02	Khande Navami	0	NaN	1	0	Thursday	October	Winter	0	1.0	0.0	0.142857	2014-10-02
268	2015-09-24	Bakri ID	1	NaN	1	0	Thursday	September	Monsoon	0	0.0	1.0	0.142857	2015-09-24
269	2015-09-24	Ganesh Festival Day8	0	NaN	1	0	Thursday	September	Monsoon	0	1.0	0.0	0.142857	2015-09-24

Rows with missing date: 14

	date	event_name	holiday_ind	remarks	Is_Event_Day	Festival_Indicator	Day_of_Week	Month_Name	Season	Is_Weekend	Holiday_Ind_Lag_1_Day	Holiday_Ind_Lag_7_Day	Holiday_Ind_Rolling_Mean_7_Day	date_parsed
628	NaN	Unlock 1.0/Mission Begin Again	1	Movement of individuals to remain strictly pro...	1	0	NaN	NaN	NaN	0	1.0	0.0	0.857143	NaT
629	NaN	Unlock 2.0/Mission Begin Again	1	NaN	1	0	NaN	NaN	NaN	0	1.0	1.0	0.857143	NaT
630	NaN	Unlock 3.0	1	Malls and market complexes without theatres, f...	1	0	NaN	NaN	NaN	0	1.0	1.0	0.857143	NaT
631	NaN	Unlock 4.0	1	Easing of restrictions, including lifting rest...	1	0	NaN	NaN	NaN	0	1.0	1.0	0.857143	NaT
632	NaN	Unlock 5.0	1	Easing of restrictions aiming to ease the flow...	1	0	NaN	NaN	NaN	0	1.0	1.0	0.857143	NaT
633	NaN	Unlock 6.0	1	NaN	1	0	NaN	NaN	NaN	0	1.0	0.0	1.000000	NaT
634	NaN	Extremely Heavy Rainfall (>100 mm)	0	NaN	1	0	NaN	NaN	NaN	0	1.0	1.0	0.857143	NaT
636	NaN	Partial Work From Home (WFH)	0	20-50% attendance in govt offices, Work from H...	1	0	NaN	NaN	NaN	0	0.0	1.0	0.571429	NaT
637	NaN	Extremely Heavy Rainfall (>100 mm)	0	NaN	1	0	NaN	NaN	NaN	0	0.0	1.0	0.428571	NaT
638	NaN	Night Curfew	0	Night curfew from 11:00 PM to 6:00 AM	1	0	NaN	NaN	NaN	0	0.0	1.0	0.285714	NaT
641	NaN	Lockdown	0	All the Public transport including Railways, B...	1	0	NaN	NaN	NaN	0	0.0	0.0	0.000000	NaT
642	NaN	Lockdown Extended	0	NaN	1	0	NaN	NaN	NaN	0	0.0	0.0	0.000000	NaT
643	NaN	Lockdown Extended	0	NaN	1	0	NaN	NaN	NaN	0	0.0	0.0	0.000000	NaT
644	NaN	Lockdown Extended	0	NaN	1	0	NaN	NaN	NaN	0	0.0	0.0	0.000000	NaT

Unique holiday_ind values: [1 0]
Non-empty remarks: 32 / 631

```
Backtesting Results (Actual vs. Predicted for the Test Set):
```

Date	Actual_Holiday_Ind_Rolling_Mean_7_Day	Predicted_Holiday_Ind_Rolling_Mean_7_Day_0.67_Quantile
2019-10-07	0.142857	0.373930
2019-10-08	0.142857	0.375623
2019-10-21	0.285714	0.445573
2019-10-25	0.285714	0.438198
2019-10-27	0.428571	0.415526

```
Calculated Penalty (Out-of-sample Pinball Loss for 0.67 quantile): 0.07197446854525098
```

```python
import numpy as np

# Make predictions on the test set
y_pred_quantile_test = lgbm_quantile_model.predict(X_test)

# Calculate pinball loss on the test set
def calculate_pinball_loss(y_true, y_pred, alpha):
    error = y_true - y_pred
    loss = np.maximum(alpha * error, (alpha - 1) * error)
    return np.mean(loss)

pinball_loss_test = calculate_pinball_loss(y_test, y_pred_quantile_test, quantile_alpha)

print(f"Out-of-sample Pinball Loss for {quantile_alpha} quantile on test set: {pinball_loss_test}")
```

```
Out-of-sample Pinball Loss for 0.67 quantile on test set: 0.07197446854525098
```

```python
import numpy as np

# 1. Calculate Forecast bias %
# Using the formula: (sum(y_pred_quantile_test) - sum(y_test)) / sum(y_test) * 100
# This approach handles potential zero values in y_test more robustly than element-wise division
forecast_bias_percentage = (np.sum(y_pred_quantile_test) - np.sum(y_test)) / np.sum(y_test) * 100

# 2. Calculate 95th percentile absolute deviation
absolute_deviations = np.abs(y_test - y_pred_quantile_test)
percentile_95_absolute_deviation = np.percentile(absolute_deviations, 95)

# 3. Print both calculated metrics
print(f"Forecast Bias %: {forecast_bias_percentage:.2f}%")
print(f"95th Percentile Absolute Deviation: {percentile_95_absolute_deviation:.4f}")
```

```
Forecast Bias %: 16.80%
95th Percentile Absolute Deviation: 0.4005
```

```python
train_size = total_samples - test_size
test_df = pd.DataFrame(index=y_test.index)
test_df['actual'] = y_test.values
test_df['forecast'] = y_pred_quantile_test
```

```python
test_df['penalty'] = np.where(
    test_df['actual'] > test_df['forecast'],
    4 * (test_df['actual'] - test_df['forecast']),
    2 * (test_df['forecast'] - test_df['actual'])
)

total_penalty = test_df['penalty'].sum()
print("Total Deviation Penalty ₹:", total_penalty)
```

```
Total Deviation Penalty ₹: 51.58128976197779
```